Shubham Gavhane

Product Engineer | GenAI | Full-Stack

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Professional Summary

Product Engineer with **4+ years of experience** building **full-stack and GenAI products** from concept to production. I specialize in developing AI-powered applications across the **backend, frontend, and AI layers**, with hands-on experience in LLM agents, RAG, Domain Knowledge Graphs, document intelligence, microservices, and scalable application architecture. I enjoy solving complex product and engineering problems and taking **end-to-end ownership** of systems that deliver measurable business impact. Currently seeking **Product Engineer, GenAI Engineer, or AI Full-Stack roles** where I can build production AI products and take ownership of meaningful engineering problems.

Experience

- **Intellect Design Arena Ltd.** Pune, India
  Product Engineer *September 2025 - Present*
  - Took an end-to-end **Trade Finance automation platform** from POC to **production within 6 months** using **React, FastAPI, LLMs, and microservices**, automating **Trade Finance -compliance and scrutiny workflows**.
  - Optimized the **end-to-end Trade Finance workflow** to reduce processing time from **8 minutes to 4 minutes** through application, microservices, and LLM workflow optimization.
  - Scaled the platform from a legacy capacity of **2–8 concurrent users to 50 users** with the same pod configuration, and enabled **120 concurrent users** through load-balancer-based scaling.
  - Built **30-40 LLM agents**, including **RAG agents**, and integrated **Domain Knowledge Graphs (DKG)** across workflows to automate Trade Finance processes; also developed an **OCRmyPDF**-based document pipeline with source traceability.
  - Developed **i3D Expert**, a **DKG-powered multi-agent document intelligence system** with **agent orchestration, tool/function calling, and specialized agents** for cross-domain document classification and key-value extraction, delivering consistent and traceable AI outputs.

- **LTIMindtree Ltd.** Bangalore, India
  Software Developer *July 2022 - September 2025*
  - Developed an **AI-powered ticket resolution agent** using **Flask, LLMs, Spring Boot, Vue.js, and ArangoDB**, reducing resolution time by **30%** and improving operational efficiency by **40%**.
  - Built a scalable **full-stack MLOps platform** using **Flask, Vue.js, Spring Boot, PostgreSQL, and Gitea** to automate ML model development, lifecycle management, and Kubernetes-based deployment.
  - Built custom **VS Code extensions** using **JavaScript and Flask** to enforce MLOps standards, validate project structures, and perform permission checks, improving developer productivity and security.
  - Developed and patented an in-house **Python ML model** for system alert correlation and automated triage, reducing resolution time by **20%** and improving operational efficiency by **15%**.

Projects

- **Online Marketplace for Collectibles**: Built AI-powered **Flask microservices** for image background removal, text extraction, product quality assessment, and recommendation systems using **OpenAI APIs**, integrated with **Spring Boot, PostgreSQL, and Vue.js**.
- **PYS - Try Before Buy**: Developed a cross-platform **Flutter** application for virtual clothing try-on, using **Firebase** for real-time data storage and user management.

Recognition & Patent

- **Spotlight Award**: Recognized for developing the Trade Finance application using a DKG-based approach.
- **Einstein Award**: Recognized for contributing to the team's success in making PRIKAR an innovative and successful solution.
- **Technical Subject Matter Expert**: Conducted a 3-day technical workshop for trainee employees.
- **Patent**: Developed and patented an in-house ML model for intelligent system alert correlation and automated triage.

Technical Skills

- **AI / GenAI**: LLMs, LLM Agents, RAG, Domain Knowledge Graphs (DKG), Prompt Engineering, Document Intelligence, OCR, Source Validation
- **Backend**: Python, FastAPI, Flask, Spring Boot, REST APIs, Microservices
- **Frontend**: React.js, Vue.js, Quasar, JavaScript, Node.js, HTML/CSS
- **Data**: PostgreSQL, MySQL, ArangoDB
- **DevOps & Engineering**: Docker, Kubernetes, Kubernetes CRDs, Git, Azure, CI/CD, System Design, Scalable Architecture, Unit Testing

Education

- **University of Mumbai** Mumbai, India
  Bachelor of Engineering in Computer Engineering, CGPA: 9.02/10.0 *Aug 2018 - May 2022*